

Does Agent Configuration Matter? An Empirical Study of Process-Level Quality Adaptation in Multi-Agent Requirements Engineering^{*}

Haowei Cheng¹[0009-0008-2265-6437]^{**}, Milhan Kim¹[0000-0003-0785-5899],
Bowen Jiang²[0009-0001-8168-0866], Weixing Zhang²[0000-0003-2890-6034],
Yuhong Fu³[0000-0003-2093-2326], Truong Vinh Truong
Duy⁴[0000-0001-8000-2214], Phan Thi Huyen Thanh⁴[0009-0007-1781-4827],
Foutse Khomh⁵[0000-0002-5704-4173], Nobukazu
Yoshioka⁶[0000-0002-1986-5675], Naoyasu Ubayashi¹[0000-0003-1578-7195],
and Hironori Washizaki¹[0000-0002-1417-9879]

¹ Waseda University, Tokyo, Japan haowei.cheng@fuji.waseda.jp

² Karlsruhe Institute of Technology, Karlsruhe, Germany

³ Adelaide University, Mawson Lakes, Australia

⁴ Aisin Corporation, Tokyo, Japan

⁵ Polytechnique Montréal, Montréal, Canada

⁶ QAML Inc., Tokyo, Japan

Abstract. Multi-agent Requirements Engineering (RE) frameworks encode an implicit process decision regarding which quality-specialized agents should be activated for a given project. However, this decision is currently made statically, creating the risk of both under-covering domain-critical quality dimensions and over-activating irrelevant agents, thereby introducing unnecessary conflicts and noise into the review process. We frame this *agent configuration* as a process-improvement decision and compare fixed-subset, full-activation, manual expert, and automated standards-driven configuration across eight case studies spanning safety-critical, financial, medical, energy, and information-system domains. The automated process combines ISO/IEC 25010 relevance scoring, deterministic domain-regulatory mapping, and project-level constraint extraction. Domain Fit Score varies $3\times$ across processes, while Conflict Noise Rate falls from 69% under full activation to 17% under automated configuration; the latter achieves 96% of expert domain fit on five primary cases and significantly outperforms both static alternatives ($p = 0.008$), generalizing across five LLM backbones and a range of thresholds. Industrial validation on an automotive perception system with an independent expert ground truth reaches 92% of expert-level domain fit with zero conflict noise. A blinded human evaluation

^{*} Replication package: <https://github.com/haowei614/configRE-code>

^{**} Corresponding author.

corroborates these rankings ($W = 0.911$, $p = 0.012$). Agent configuration materially affects RE outcomes and can be audited through standards traceability; standards-driven adaptation improves multi-agent RE across heterogeneous portfolios.

Keywords: Requirements engineering · software process improvement · process tailoring · multi-agent systems · large language models · ISO/IEC 25010 · empirical study · domain adaptation

1 Introduction

Modern systems must simultaneously satisfy competing non-functional requirements (safety, efficiency, sustainability, trustworthiness, regulatory compliance), and as complexity scales, manually balancing these dimensions becomes increasingly challenging [1,2,3].

Large language models (LLMs) have enabled multi-agent frameworks that decompose requirements analysis across specialized agents, each advocating a distinct quality objective [4,5,6,7]. Using such a framework introduces a process decision that is easy to overlook: *which quality-specialized agents should participate in the negotiation?* As in collaborative RE, where the represented stakeholder perspectives shape the resulting requirements, the agent set determines which quality concerns are voiced, negotiated, and reconciled; its consequences propagate into the RE product.

We refer to this process step as *agent configuration*. Current multi-agent RE frameworks execute it with a one-size-fits-all process: a fixed set of quality-specialized agents is activated regardless of the target project’s domain, scale, or regulatory context [4,5]. This contradicts a core tenet of process engineering, namely Pohl’s process-context fit principle [1] and CMMI tailoring guidance [8], which holds that processes should be adapted to project context rather than applied uniformly. A static process introduces two systematic risks: *under-coverage*, where domain-critical dimensions absent from the fixed set are neglected (e.g., an autonomous-driving perception module needs real-time performance and functional-safety reasoning under ISO 26262, which a generic set omits); and *over-activation*, where agents irrelevant to the domain inject unnecessary conflict proposals, raising cost and degrading negotiation quality.

This tension surfaces concretely in industrial settings, where the same RE framework must serve diverse project domains. The contrast between automotive safety-critical systems and information management systems motivated the present study in collaboration with an automotive supplier (Aisin). A default five-agent set partially fits an *autonomous-driving* project but misses Performance and FunctionalSafety; for a *library management system*, the mismatch is total—its primary concerns (Usability, Compatibility, Maintainability) are absent, yielding a Domain Fit Score (DFS) of 0.00, while irrelevant agents inject pure noise (Section 5). Despite these intuitive stakes, no prior work has empirically investigated the extent to which agent configuration affects multi-agent RE outcomes. This raises two concrete questions: (1) *Does the choice of which*

agents to activate actually matter for product quality? and (2) *Can this process step be automated without domain expertise?*

This paper presents a controlled empirical study answering these questions. We operationalize agent configuration as selecting a subset from a pool of 15 quality-specialized agents grounded in ISO/IEC 25010 [9] and domain-specific standards. Across eight case studies, we compare four configuration processes: fixed-subset, full activation, manual expert selection, and automated standards-driven selection. To enable automated selection, we develop a three-tier process combining LLM-based relevance scoring (Tier 1), deterministic domain-regulatory mapping (Tier 2), and project-level constraint extraction (Tier 3), which serves as the apparatus for evaluating whether principled automation is feasible.

The main contributions are: (1) **empirical evidence that agent configuration is consequential**—DFS varies $3\times$ (0.26–0.77) and conflict noise from 15% to 69%; (2) automated adaptation reaches 96% of manual expert quality without domain expertise; (3) a reproducible three-tier configuration process grounded in ISO/IEC 25010 and domain regulations; and (4) **industrial validation** on a real automotive system with an independent expert ground truth (92% expert DFS, zero noise).

2 Methodology

2.1 Study Design Overview

We introduce a Phase 0 (Domain-Adaptive Agent Configuration) preceding the standard negotiation pipeline. As shown in Figure 1, Phase 0 takes the natural-language project description and produces an activated set $AG^* \subseteq AG_{\text{pool}}$ (15 agents), passed to Phase 1, which instantiates only the selected agents; Phase 2’s ring-based pairwise negotiation adapts automatically by deriving pairs from the Phase 1 output, and Phases 3–5 are unchanged. Varying the Phase 0 strategy thus compares agent-selection approaches under otherwise identical conditions. Negotiation cost scales as $O(|AG^*|)$, so a typical $|AG^*| = 4\text{--}7$ keeps cost manageable.

2.2 Agent Pool Design

The agent pool AG_{pool} comprises 15 quality-specialized agents (Table 1). Ten derive from ISO/IEC 25010:2023 (splitting *Performance Efficiency* into separate Performance and Efficiency agents; excluding *Functional Suitability*, assumed captured in initial requirements). Five correspond to domain-specific standards. The pool is *flat*: the three tiers in Section 2.3 are *activation mechanisms*, any of which may select any agent.

Each agent’s prompt template, role definition, and JSON output schema are detailed in the replication package.

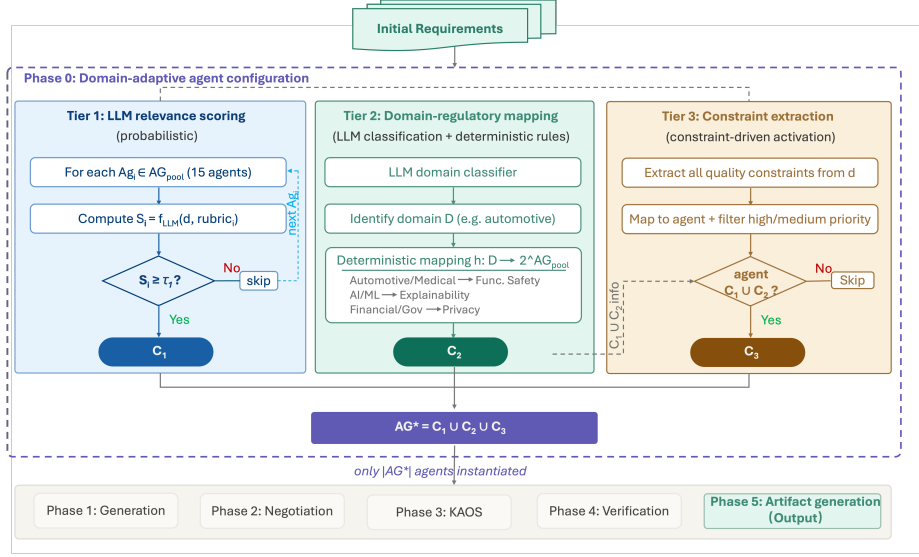


Fig. 1. Experimental setup. Phase 0 determines agent configuration via three mechanisms: Tier 1 (LLM relevance scoring), Tier 2 (deterministic domain mapping), and Tier 3 (constraint extraction), producing candidate sets C_1, C_2, C_3 ; the activated set $AG^* = C_1 \cup C_2 \cup C_3$ is passed to Phase 1. AG_{pool} = full pool of 15 quality-specialized agents; C_1, C_2, C_3 = candidate sets from Tiers 1–3 respectively; $AG^* = C_1 \cup C_2 \cup C_3$ = final activated agent set passed to Phase 1. Phases 1–5 follow the standard QUARE/ArgRE pipeline [4,5].

2.3 Three-Tier Agent Selection Taxonomy

The three tiers are *activation mechanisms with complementary reliability guarantees*, not disjoint agent sets: any agent in AG_{pool} can be activated by any tier, and the tiers are *additive*, yielding the final set $AG^* = C_1 \cup C_2 \cup C_3$ with monotonic coverage.

Tier 1 (LLM relevance scoring, probabilistic). For each agent Ag_i , an LLM computes a zero-shot relevance score $s_i^{(1)} = f_{LLM}(d, rubric_i) \in [0, 1]$ from the project description d , using a five-level ordinal rubric anchored at 1.0 (critical), 0.8 (major), 0.5 (secondary), 0.3 (minor), 0.0 (irrelevant) [10]. Agents with $s_i^{(1)} \geq \tau_1$ (default $\tau_1 = 0.6$) form C_1 . This gives broad coverage but is probabilistic: scoring variance may place regulatory-critical agents below the threshold. Tier 2 addresses this risk.

Tier 2 (domain-regulatory mapping, deterministic). Tier 2 guards regulatory-mandatory agents. It classifies the project’s domain via an LLM, then applies a deterministic mapping h to obtain $C_2 = \bigcup_{d \in D} h(d)$. The current mapping covers three regimes: automotive/medical activate FunctionalSafety (ISO 26262 / IEC 62304); AI/ML activate Explainability (EU AI Act); finan-

Table 1. Quality agent pool ($|AG_{\text{pool}}| = 15$). Each agent advocates a distinct quality concern grounded in the listed standard. Ten agents derive from ISO/IEC 25010:2023; the remaining five correspond to domain-specific standards. $|AG_{\text{pool}}|$ denotes the total pool size.

Agent	Standard	Sub-focus
Safety	ISO 25010	Hazard prevention, risk mitigation
Performance	ISO 25010	Response time, throughput, capacity
Efficiency	ISO 25010	Resource utilization, energy usage
Reliability	ISO 25010	Fault tolerance, recoverability
Usability	ISO 25010	Learnability, error protection
Security	ISO 25010	Authentication, access control
Trustworthiness	ISO 25010	Privacy, data protection, trust
Maintainability	ISO 25010	Modularity, testability
Compatibility	ISO 25010	Interoperability, co-existence
Flexibility	ISO 25010	Adaptability, replaceability
Functional Safety	ISO 26262	ASIL levels, hazard analysis
Explainability	EU AI Act	Model transparency, interpretability
Privacy	GDPR	Consent, data subject rights
Green	ISO 14001	Energy efficiency, carbon footprint
Responsibility	IEEE 7000	Ethical accountability, compliance

cial/government activate Privacy (General Data Protection Regulation, GDPR). It is extensible by design.

Tier 3 (project-level constraint extraction, constraint-driven). Tier 3 closes a granularity gap: dimensions mentioned only locally may receive low holistic scores. It extracts explicit high/medium-priority quality constraints and activates their mapped agents when not already in $C_1 \cup C_2$: $C_3 = \text{extract}_{\text{LLM}}(d, AG_{\text{pool}} \setminus (C_1 \cup C_2))$.

3 Related Work

3.1 LLM-based and Multi-Agent Requirements Engineering

LLMs have been applied to diverse RE tasks [11,12,13,14,15], yet systematic mappings find that multi-agent architectures remain a minority [16] and adaptive-configuration approaches are largely absent [17]. Multi-agent RE frameworks follow two design philosophies: *task-specialized* agents (MARE [6], iReDev [7]) and *quality-specialized* agents with dialectical negotiation (QUARE [4], ArgRE [5]). In all cases, however, the agent set is fixed at design time; the effect of this choice on downstream RE product quality remains unstudied.

3.2 Dynamic Team Composition in Multi-Agent LLM Systems

General-purpose multi-agent frameworks such as AutoGen [18], MetaGPT [19], and CrewAI support dynamic team assembly with role-assignment heuristics.

Their composition mechanisms, however, target general software-engineering tasks and lack RE-specific quality taxonomies and domain-regulatory constraints, making them unsuitable for standards-driven agent selection in RE.

3.3 Software Process Improvement and Tailoring

Process tailoring, i.e. adapting a reference process to project context, is an established principle in software process improvement (SPI) frameworks such as CMMI [8] and ISO/IEC 15504 [20], driven by domain, criticality, and regulatory context [21,22]. We apply tailoring to a new target—the composition of quality-specialized agents in multi-agent RE—and measure its product-quality impact empirically.

3.4 Quality Models and Standards

Our agent pool derives ten agents from ISO/IEC 25010:2023 [9] and five from domain-specific standards (ISO 26262 [23], EU AI Act [24], GDPR, ISO 14001) whose regulatory constraints a static agent set cannot anticipate. Tier 2’s deterministic mapping guarantees their activation regardless of Tier 1 scoring.

4 Experimental Setup

4.1 Research Questions

- **RQ1 (Configuration Effect):** How does the agent configuration *process* affect domain fit, Conflict Noise Rate (CNR), and semantic preservation?
- **RQ2 (Dynamic vs. Static):** Does *adaptive* configuration *as a process step* outperform fixed-subset and full-activation processes across diverse project domains?
- **RQ3 (Selection Accuracy):** How accurately does the automated selection mechanism identify domain-relevant quality dimensions compared to expert judgment?
- **RQ4 (Robustness and Generalizability):** Do these findings generalize to additional external domains, and do they remain robust to the Tier 1 threshold τ_1 and to the choice of LLM backbone?

4.2 Case Studies

We evaluate on eight case studies (Table 2): five from prior work [4] and three external cases (EHR, Smart Grid, Loan Approval) constructed from public sources [25] and anchored to applicable standards (HIPAA/HL7 FHIR, EU AI Act, ISO 14001) to test generalization. Ground truth and rationale are in the replication package.

Table 2. Summary of case studies.

Case Study	Domain	Core Focus	Source
Autonomous Driving (AD)	Automotive	Safety	QUARE
ATM System	Finance	Security	QUARE
Library System	Management	Scalability	QUARE
RollCall System	Management	Functional	QUARE
Bookkeeping System	Finance	Data Integrity	QUARE
Electronic Health Records	Medical	Privacy/Safety	PROMISE NFR
Smart Grid	Energy/IoT	Reliability/Green	Public sources
Loan Approval	Finance/AI	Explainability	PROMISE NFR

4.3 Compared Configurations

We compare four configurations: **Fixed-5** (QUARE/ArgRE baseline: Safety, Efficiency, Green, Trustworthiness, Responsibility, uniformly activated); **Full-15** (all 15 agents); **Domain-opt** (a manually selected set of ~ 6 domain-relevant agents per case, an expert upper-bound reference); and **Phase0-Auto** (automated three-tier selection, Section 2).

4.4 Implementation Details

All experiments use `gpt-4o-mini-2024-07-18` (temperature 0.7), $\tau_1 = 0.6$, negotiation capped at three rounds per pair, and three seeds (101, 202, 303) averaged, built on the OpenReBench platform [4].

4.5 Evaluation Metrics

Against the expert ground truth AG_{GT} , we compute Agent Selection Precision $ASP = |AG^* \cap AG_{GT}|/|AG^*|$ and Recall $ASR = |AG^* \cap AG_{GT}|/|AG_{GT}|$, and report their harmonic mean, the Domain Fit Score $DFS = 2 ASP ASR / (ASP + ASR)$. The Conflict Noise Rate (CNR) is the proportion of detected conflicts involving at least one agent not in AG_{GT} ; by convention, we report $CNR = 0$ when no conflicts are detected. Semantic preservation is BERTScore F1 between Phase 3 and Phase 1 outputs via pairwise optimal bipartite matching, following QUARE and ArgRE.

4.6 Expert Ground Truth

The first author identified the relevant quality dimensions for each case, producing AG_{GT} with documented rationale. Validation escalates in independence: the primary cases were checked by one independent annotator ($\kappa = 0.715$), the external cases by three independent annotators under a blinded protocol (Fleiss’ $\kappa = 0.820$), and the industrial case uses ground truth supplied by an independent Aisin domain expert. Annotator backgrounds span PhD, Master’s, and undergraduate levels in software engineering (1–5 years of RE experience); detailed profiles and item-level disagreements are in the replication package.

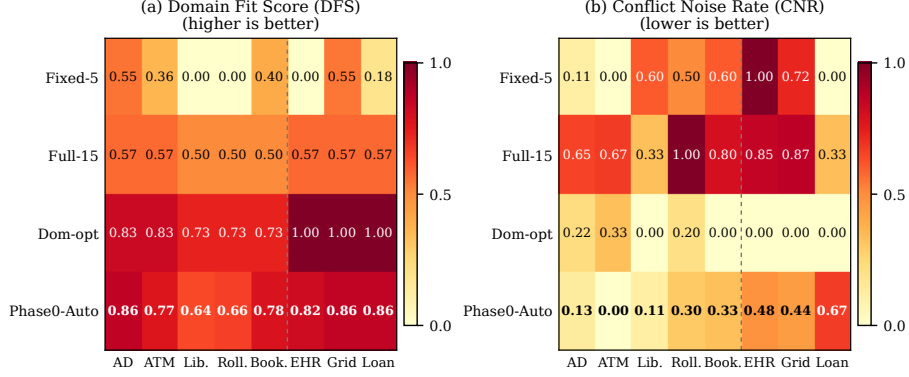


Fig. 2. Case-level Domain Fit Score (DFS; higher is better) and Conflict Noise Rate (CNR; lower is better) for the four configuration processes. Each cell reports the metric value averaged over 3 seeds. Darker cells indicate larger values. The gray dashed line separates the five primary cases (left) from the three external validation cases (right). For the external cases, Domain-opt coincides with the operational ground truth by construction, so DFS = 1.00 and CNR = 0. AD = Autonomous Driving, ATM = ATM System, Lib. = Library System, Roll. = RollCall System, Book. = Bookkeeping System, EHR = Electronic Health Records, Grid = Smart Grid, Loan = Loan Approval.

4.7 Human Evaluation Protocol

To test whether the automated metric differences are reflected in perceived quality, we conduct a blinded human evaluation of Phase 5 outputs (scoping in Section 8). For four cases (EHR, Smart Grid, Loan Approval, W-Mobility), specifications from all configurations are rendered to uniform Markdown, randomly relabeled A–D (16 documents), and scored by three software-engineering evaluators at PhD, Master’s, and undergraduate levels (detailed profiles in the replication package) on four Likert-1–5 dimensions (*Completeness*, *Relevance*, *Actionability*, *Consistency*) after a calibration session. We report ordinal Krippendorff’s α and Kendall’s W .

5 Results and Analysis

5.1 RQ1: Agent Configuration Effect

Configuration strongly affects domain fit. Average DFS ranges $3\times$ from 0.26 (Fixed-5) to 0.77 (Domain-opt), collapsing to 0.00 on Library and RollCall where the default agents have zero overlap with the needed dimensions. Figure 2 shows that this is not an average driven by one case: Phase0-Auto exceeds both static configurations on DFS in all eight cases. **Over-activation injects conflict noise:** Full-15 averages CNR = 69%, i.e. over two-thirds of conflicts involve a domain-irrelevant agent. Phase0-Auto keeps CNR at 0.00–0.33 on the

Table 3. Cross-domain comparison of agent configuration strategies (averaged over 3 seeds per configuration). DFS = Domain Fit Score (higher is better), CNR = Conflict Noise Rate (lower is better), BERT = BERTScore F1 (semantic preservation between Phase 3 and Phase 1 outputs), Tok(K) = total token consumption in thousands. Avg5 = average over the five primary cases; Avg8 = average over all eight cases. Cases left of the vertical rule are from the primary study; cases to its right are external validation domains.

Config	Metric	AD	ATM	Lib.	Roll.	Book.	EHR	Grid	Loan	Avg ₅	Avg ₈
Fixed-5	DFS $\uparrow$	0.55	0.36	0.00	0.00	0.40	0.00	0.55	0.18	0.26	0.26
	CNR $\downarrow$	0.11	0.00	0.60	0.50	0.60	1.00	0.72	0.00	0.36	0.44
	BERT	.947	.954	.948	.945	.948	.946	.947	.947	.948	.948
	Tok(K)	20	18	18	16	16	19	16	14	18	17
Full-15	DFS $\uparrow$	0.57	0.57	0.50	0.50	0.50	0.57	0.57	0.57	0.53	0.54
	CNR $\downarrow$	0.65	0.67	0.33	1.00	0.80	0.85	0.87	0.33	0.69	0.69
	BERT	.954	.959	.954	.952	.954	.955	.955	.953	.955	.955
	Tok(K)	62	58	53	47	47	57	63	46	53	54
Dom-opt	DFS $\uparrow$	0.83	0.83	0.73	0.73	0.73	1.00	1.00	1.00	0.77	0.86
	CNR $\downarrow$	0.22	0.33	0.00	0.20	0.00	0.00	0.00	0.00	0.15	0.09
	BERT	.954	.959	.954	.950	.953	.959	.956	.957	.954	.955
	Tok(K)	23	21	19	23	17	22	21	19	21	21
Ph0-Auto	DFS $\uparrow$	0.86	0.77	0.64	0.66	0.78	0.82	0.86	0.86	0.74	0.78
	CNR $\downarrow$	0.13	0.00	0.11	0.30	0.33	0.48	0.44	0.67	0.17	0.31
	BERT	.954	.960	.952	.951	.955	.957	.954	.953	.954	.955
	Tok(K)	36	34	27	33	32	42	37	38	32	35

Agent counts: Fixed-5 = 5, Full-15 = 15, Dom-opt $\approx$ 6, Ph0-Auto = 4.3–8.7 (mean 6.5); counts are deterministic except for Phase0-Auto. For external cases, Domain-opt = ground truth by construction (DFS = 1.00, CNR = 0). Wilcoxon: Ph0-Auto vs. Fixed-5/Full-15: $p=0.008$ at $n=8$ ($W=0$, $\delta=1.0$).

primary cases, though the 0.44–0.67 range on external cases shows that unfamiliar domains still reduce selection precision. **Semantic preservation is robust:** BERTScore F1 stays at 0.948–0.955, as the KAOS integration phases normalize output quality regardless of the upstream agent set.

5.2 RQ2: Dynamic vs. Static Configuration

Phase0-Auto outperforms Fixed-5 on every primary case (average DFS 0.74 vs. 0.26, +0.48; largest gains on Library 0.64 vs. 0.00 and RollCall 0.66 vs. 0.00) and beats Full-15 on every case as well (+0.21 DFS, −52pp noise: 0.17 vs. 0.69). It approaches manual expert selection (96% of Domain-opt’s 0.77 on the five primary cases; 85% on the external cases where Domain-opt = AG_{GT} by construction) and even *exceeds* it on AD (0.86 vs. 0.83), where Tier 1 correctly flagged Performance that the manual set omitted. Phase0-Auto beats Full-15 on DFS while using 60% of its token budget (32K vs. 53K).

5.3 RQ3: Selection Accuracy

Per-case selection metrics confirm the DFS ordering in Table 3. On the five primary cases, Phase0-Auto attains ASP/ASR/DFS of .89/.83/.86 (AD), .71/.83/.77 (ATM), .70/.60/.64 (Library), .61/.73/.66 (RollCall), and .76/.80/.78 (Book-keeping), versus Fixed-5’s collapse to 0 on Library and RollCall and Full-15’s uniformly low precision (ASP $\approx$ 0.33–0.40 at ASR = 1.0).

Tier Contribution Analysis A cumulative ablation (five primary cases $\times$ 3 seeds) shows complementary, diminishing contributions. Tier 1 alone reaches DFS = 0.72 with 4.2 agents; Tier 2 raises it to 0.75 (AD 0.66 $\rightarrow$ 0.79 through activation of FunctionalSafety), and Tier 3 adds refinement (AD 0.79 $\rightarrow$ 0.86). Despite case-level fluctuation, average CNR falls monotonically (0.21 $\rightarrow$ 0.18 $\rightarrow$ 0.17).

Failure modes. Two error categories trade off through τ_1 . (1) *False activations* reduce ASP when Tier 1 scores marginally relevant agents above threshold—e.g., Trustworthiness scores 0.8 for Library because its broad descriptor (“data protection, trust”) overlaps with generic system features. (2) *Missed agents* reduce ASR when implicit requirements are not lexically present—e.g., Compatibility scores 0.3 for Library because the description omits interoperability concerns that a domain expert would infer. Tractable mitigations include calibrated rubric descriptors with domain-specific vocabulary and post-hoc pruning of semantically overlapping agents. The threshold sensitivity analysis confirms that $\tau_1 \in [0.6, 0.8]$ provides a stable operating region balancing both error types.

5.4 RQ4: Robustness and Generalizability

Generalization to External Domains The three external cases (Table 3, right) confirm the effect persists: Phase0-Auto averages DFS = 0.85 versus 0.24 (Fixed-5) and 0.57 (Full-15), with Fixed-5 again collapsing on domain-divergent projects (DFS = 0.00 for EHR). Here Domain-opt coincided with the operational ground truth (DFS = 1.00), so it acts as an idealized upper bound; Phase0-Auto attains 85% of this bound without domain expertise. The ground truth was subsequently assessed through a three-annotator blinded validation (Section 8). Phase0-Auto activates more agents on these newer domains (8.2 vs. 5.7), raising CNR to 0.53; 60% of false activations originate in Tier 1’s broad rubric descriptors, leaving room to tighten Tier 1 precision on unfamiliar domains.

Pooled significance ($n = 8$). Combining all eight cases, Wilcoxon signed-rank tests confirm Phase0-Auto significantly outperforms both Fixed-5 and Full-15 ($W = 0$, $p = 0.008$, Cliff’s $\delta = 1.0$; 8/8 cases favor Phase0-Auto). Unlike the $n = 5$ analysis bounded at $p = 0.0625$, the larger sample crosses $\alpha = 0.05$ while preserving a maximal effect size.

Threshold Sensitivity (τ_1) Figure 3 shows the trade-off induced by varying τ_1 from 0.3 to 0.9. Raising τ_1 from 0.3 to 0.6 reduces the mean activated set

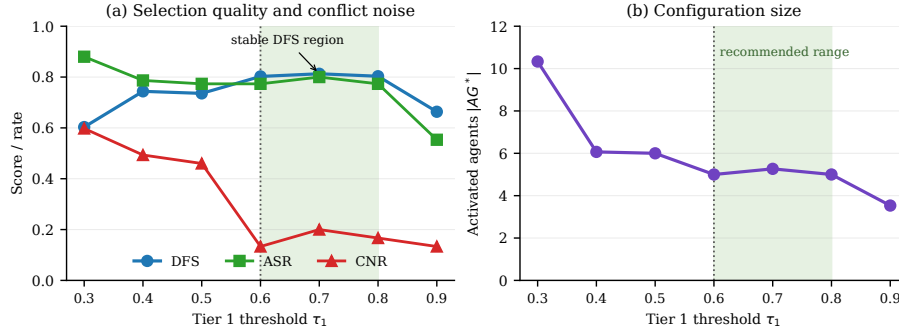


Fig. 3. Tier 1 relevance-score threshold (τ_1) sensitivity, averaged over the five primary cases and 3 seeds. τ_1 controls which agents are activated by Tier 1: agents with relevance score $\geq \tau_1$ enter candidate set C_1 . Panel (a): DFS = Domain Fit Score, ASR = Agent Selection Recall, CNR = Conflict Noise Rate. The shaded region marks the stable operating range $\tau_1 \in [0.6, 0.8]$; the dotted line marks the default $\tau_1 = 0.6$. Panel (b): mean number of activated agents $|AG^*|$; the shaded region marks the recommended range.

from $|AG^*| = 10.3$ to 5.0 and CNR from 0.60 to 0.13, while DFS increases from 0.60 to 0.80. DFS is then nearly flat across $\tau_1 \in [0.6, 0.8]$ (0.80–0.81, $\text{CNR} \leq 0.20$), confirming that the default $\tau_1 = 0.6$ is well-calibrated. At $\tau_1 = 0.9$, the smaller set under-covers relevant dimensions (ASR drops to 0.55, DFS to 0.66). The plateau across three adjacent values indicates the mechanism is not brittle to the exact choice of τ_1 .

Cross-Model Validation We re-ran Phase 0 selection with GPT-4o-mini, Claude Sonnet 4.5, Llama 3.1 8B, Qwen2.5 7B, and Gemma 2 9B. Phase0-Auto preserves the ordering for every backbone (Avg₈ DFS 0.66–0.78 versus the model-independent Fixed-5 reference of 0.26; $p = 0.008$ for each). The open-weight models match or exceed Claude Sonnet 4.5, whose larger activated sets favor recall over precision. Full per-model results are in the replication package.

5.5 Human Evaluation of Downstream Requirement Quality

The blinded human evaluation (Section 4.7) serves as corroborative ranking evidence: it tests whether evaluators’ perceived quality rankings align with the automated DFS ordering. Table 4 reports mean Likert scores per configuration, averaged over four cases and three evaluators (48 document-level ratings).

All three evaluators, despite spanning PhD, Master’s, and undergraduate training levels, produced strongly convergent configuration rankings (Kendall’s $W = 0.911$): Domain-opt ranks first and Phase0-Auto second, with the two static configurations last. This top-two ordering and the separation from the static alternatives corroborate the automated DFS findings. A Friedman test

Table 4. Human evaluation of Phase 5 requirement specifications (mean Likert 1–5, averaged over 4 case studies and 3 evaluators). Compl. = Completeness, Relev. = Relevance, Action. = Actionability, Consist. = Consistency, each rated on a 1–5 Likert scale. Krippendorff’s α (ordinal) measures absolute inter-rater agreement; Kendall’s W measures rank-order concordance across evaluators. Absolute scores are reported for transparency; the low inter-rater agreement ($\alpha = 0.323$) limits their interpretability. The primary finding is the ranking convergence across evaluators ($W = 0.911$).

Config	Compl.	Relev.	Action.	Consist.	Avg
Fixed-5	2.50	2.83	2.17	2.67	2.54
Full-15	4.00	2.00	2.00	2.00	2.50
Domain-opt	3.92	3.75	2.83	3.25	3.44
Phase0-Auto	3.50	3.17	2.58	2.92	3.04

Krippendorff’s α (ordinal) = 0.323; Kendall’s W on configuration rankings = 0.911. Friedman $\chi^2 = 10.89$, $p = 0.012$.

confirms a significant configuration effect ($\chi^2 = 10.89$, $p = 0.012$). Mean Likert scores are directionally consistent with this ranking: Domain-opt leads at 3.44 and Phase0-Auto follows at 3.04. However, absolute-score agreement is only fair (ordinal $\alpha = 0.323$), so these values should be interpreted as indicative rather than calibrated estimates. Evaluators rank Full-15 lowest overall despite its highest Completeness score (4.00), penalizing it on Relevance, Actionability, and Consistency—a pattern consistent with the conflict noise that CNR captures.

This pattern suggests that evaluators implicitly penalize noise even without knowing which configuration produced each document: the irrelevant quality dimensions injected by over-activation dilute the specification’s focus, making it harder to extract actionable guidance. The gap between Domain-opt (3.44) and Phase0-Auto (3.04) is smaller than between Phase0-Auto and the static alternatives (2.54, 2.50), reinforcing the view that automated configuration captures most of the expert advantage.

6 Industrial Validation

To test transfer to a genuine industrial setting with an *independent expert* ground truth, we ran a case study with an automotive supplier (Aisin). The target system is an AI-based perception module for pedestrian and obstacle detection in a pedestrian-mixed campus environment, governed by the Japanese Road Transport Vehicle Act and specified-small-motorized-bicycle regulations. Its stringent resource and safety constraints make trade-offs among real-time performance, functional safety, and energy efficiency consequential. The ground truth was defined by an independent domain expert from Aisin rather than by the authors, thereby reducing potential author bias and better reflecting industrial practice. We acknowledge that this remains a single-annotator judgment; broader multi-annotator validation of the industrial ground truth is left to future work (Section 9). The expert identified seven relevant agents (Safety, Performance,

Table 5. Industrial case study (W-Mobility): configuration strategies against independent expert ground truth, averaged over 3 seeds. **Number of Agents** reports the mean count of activated agents across seeds; for Fixed-5, Full-15, and Domain-opt this count is deterministic across seeds, while for Phase0-Auto it may vary by seed, producing non-integer means. ASP = Agent Selection Precision ($|AG^* \cap AG_{GT}|/|AG^*|$), ASR = Agent Selection Recall ($|AG^* \cap AG_{GT}|/|AG_{GT}|$), DFS = Domain Fit Score (harmonic mean of ASP and ASR; higher is better), CNR = Conflict Noise Rate (proportion of conflicts involving at least one irrelevant agent; lower is better). Fixed-5 / Full-15 / Domain-opt / Phase0-Auto are the four configuration processes defined in Section 4.3. Domain-opt coincides with the expert ground truth and thus serves as an idealized upper bound (DFS = 1.00). BERTScore F1 is stable across configurations (0.950–0.956); token cost ranges from 15K (Fixed-5) to 58K (Full-15), with Phase0-Auto at 37K.

Config	Number of Agents	ASP	ASR	DFS $\uparrow$	CNR $\downarrow$
Fixed-5	5.0	0.40	0.29	0.33	0.67
Full-15	15.0	0.47	1.00	0.64	0.73
Domain-opt	7.0	1.00	1.00	1.00	0.00
Phase0-Auto	6.0	1.00	0.86	0.92	0.00

Efficiency, Reliability, Usability, FunctionalSafety, Privacy), so Domain-opt here comprises these seven. We ran all four configurations under the main-study protocol (QUARE backbone, GPT-4o-mini, three seeds).

The results (Table 5) reproduce every main-study effect on real industrial data. Fixed-5 fails to fit the domain (DFS = 0.33, CNR = 0.67); Full-15 attains perfect recall but at low precision (ASP = 0.47), the highest noise (CNR = 0.73), and $\sim 1.6\times$ the token cost of Phase0-Auto. Phase0-Auto reaches DFS = 0.92 (92% of the expert upper bound) with perfect precision (zero noise) and 63% of Full-15’s tokens. Tier 2 activated FunctionalSafety, and Tier 3 recovered Usability; Privacy was the only missed dimension. The fact that Phase0-Auto achieved perfect precision (ASP = 1.00) on an unseen industrial system, with ground truth defined entirely outside the research team, provides the strongest evidence that the three-tier mechanism transfers beyond the constructed benchmarks.

7 Discussion

Adaptive configuration is most beneficial under three conditions: (i) *domain heterogeneity*, where one framework serves domains with differing quality priorities; (ii) *coverage gaps*, where more than half a project’s primary concerns are absent from the default set (e.g., DFS = 0.00 $\rightarrow$ 0.64 for Library); and (iii) *regulatory exposure*, where ISO 26262, GDPR, or EU AI Act mandate specific agents that Tier 2 guarantees. Exhaustive activation is counterproductive: Full-15 drives CNR to 69% and nearly triples cost; ~ 5 –7 targeted agents consistently outperform.

Phase 0 adds only $\sim 5\text{K}$ tokens ($< 15\%$ of pipeline cost, under 30 s wall-clock), yielding 0.032 DFS/K-token versus 0.008 for Full-15 ($4\times$ more token-efficient) while reducing CNR by 52 percentage points at 60% of Full-15’s budget. The industrial case reinforces this: Phase0-Auto achieves $\text{DFS} = 0.92$ with zero noise at 63% of Full-15’s token cost.

Four implications follow from the results: static agent sets must be validated rather than assumed (DFS collapses to 0.00 when domains diverge); more agents do not mean better quality (Full-15 drives CNR to 69%); standards-based automation can substitute for domain expertise (96% of expert DFS on primary cases); and Phase 0 is orthogonal to the negotiation mechanism, composing with both heuristic and argumentation-based resolution for incremental adoption. These lessons are most relevant for organizations running one framework across heterogeneous portfolios under standards such as ISO 26262 and GDPR, where an auditable rationale for each quality dimension is expected.

8 Threats to Validity

Construct validity. The ground truth (AG_{GT}) was initially constructed by the first author; we escalate validation independence across study stages. Primary cases were checked by one independent annotator ($\kappa = 0.715$); disagreements concentrated on boundary agents (e.g., Green for AD, Privacy for RollCall). External cases were validated by three independent annotators (Fleiss’ $\kappa = 0.820$); disagreements were limited to boundary dimensions (e.g., Maintainability for Smart Grid). The industrial ground truth was defined entirely by an independent Aisin expert. Flipping any single boundary agent changes DFS by at most ± 0.05 ; under majority-vote relabeling the overall ranking is preserved (Domain-opt $>$ Phase0-Auto $>$ Full-15 $>$ Fixed-5). The human evaluation is scoped as corroborative ranking evidence ($W = 0.911$, $p = 0.012$); absolute-score agreement is only fair ($\alpha = 0.323$), and a larger evaluation is planned.

Internal validity. Non-determinism in Tier 1/Tier 2 is mitigated via three seeds. Temperature and backbone are held constant across configurations, so they affect absolute quality equally. The default $\tau_1 = 0.6$ is stable across $[0.6, 0.8]$; pooling all eight cases yields $p = 0.008$ (Cliff’s $\delta = 1.0$).

External validity. The three external cases derive from PROMISE NFR [25] and real standards, but may not capture field-elicited messiness; the industrial case partly offsets this. The 15-agent pool is not exhaustive. Cross-model replication across five backbones confirmed the effect for Phase 0 selection; full-pipeline cross-backbone validation remains future work.

9 Conclusion

The experiments show that agent configuration determines which quality concerns enter multi-agent RE negotiation: DFS varies $3\times$ across processes while semantic quality stays stable, and automated standards-driven selection reaches 96% of manual expert domain fit without domain expertise. The effect generalizes

across external domains, five LLM backbones, a range of thresholds ($p = 0.008$), an industrial case with independent expert ground truth (92% expert DFS, zero noise), and a blinded human evaluation ($W = 0.911$, $p = 0.012$). Agent configuration should be treated as a first-class, auditable process step in multi-agent RE. Future work includes larger-scale industrial studies with multi-annotator expert panels across additional domains (e.g., aerospace, telecommunications), a scaled human evaluation targeting at least eight evaluators per case to enable calibrated absolute-score comparisons beyond the ranking evidence reported here, integration with argumentation-based conflict resolution such as ArgRE’s Dung-framework backend, and exploration of adaptive threshold mechanisms that adjust τ_1 per domain based on historical selection accuracy.

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